Threshold Post-Quantum Signing, and the Holder Side with Longfellow

A short conversation deck

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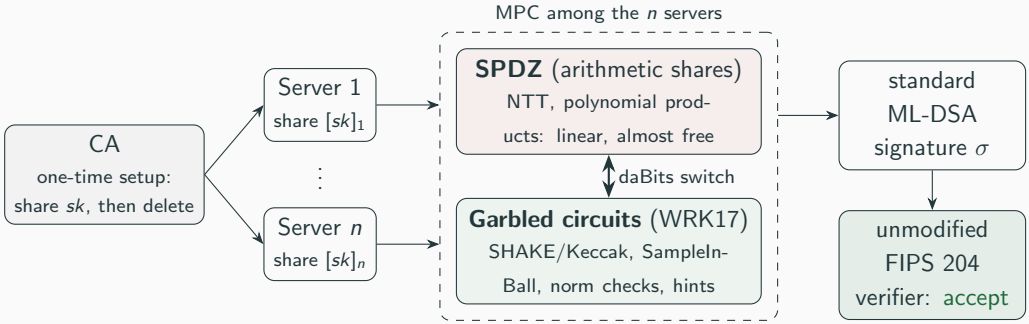
October 30, 2026

About me

- Fourth-year PhD student at Virginia Tech, applied cryptography; graduating 2028.
- Main line: **threshold signing for lattice signatures**, so that no single server holds the key and the output is a standard signature.
  - **SHIELD** (IEEE S&P 2026, co-first author): threshold ML-DSA.
  - **FLOCK** (first author, under submission): Falcon-compatible threshold signatures.
- Also ORAM: **HERO** (first author, under review), distributed ORAM with an untrusted helper.

Common thread: compute and prove things about unmodified standard signatures.

SHIELD (IEEE S&P 2026): threshold ML-DSA, standard output



- No single server ever holds the key: n-of-n, malicious security with abort (up to $n-1$ corrupted servers).
- Output is bit-identical to FIPS 204 (NIST KATs); Boolean steps compiled Verilog $\rightarrow$ Yosys $\rightarrow$ Bristol circuits.

FLOCK: Falcon-compatible threshold signing in nine rounds

Falcon's trapdoor sampler (ffSampling)
floating point; $2n$ sequential
draws, each centered on a secret

in MPC

prior threshold Falcon
tens of thousands of
rounds per signature

① **Swap the sampler**: Peikert offline/online + Zalcon + Antrag
trapdoor $\Rightarrow$ integer-only, parallel, sequential depth 2 instead of $2n$

② **Keep the secret-centered draw outside the MPC**

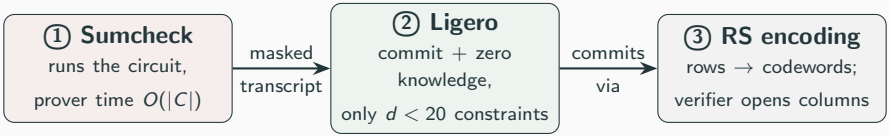
center c in secret shares → open the masked center $c + e$ → each party samples in the clear → back in MPC: linear ops only

Offline batched preprocessing + 9 online rounds; signatures pass Falcon's verifier at FLOCK's parameters ($-4/12$ forgery bits, $+9/32$ bytes)

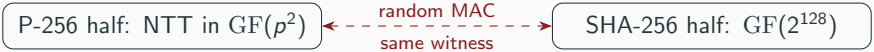
4 signers, smaller set: 25 ms on a LAN vs. 24.2 s for Garg et al. (first author, under submission).

My reading of Longfellow

Statement: I hold an mdoc signed with ECDSA P-256 + SHA-256; reveal one attribute, hide the device key

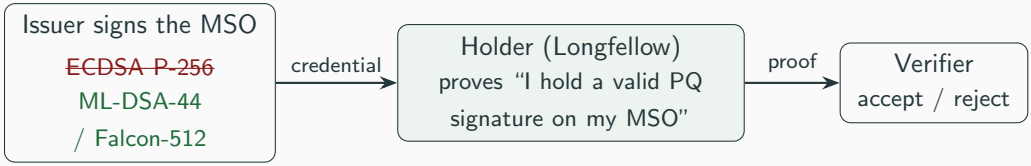


only SHA-256 is assumed: transparent, hash-based



ECDSA prover 16.7 ms on an M4; CFRG draft; part of the PQCA. Frigo & shelat, CiC vol. 3 (2026); IETF 126 slides.

What comes next: swap the issuer's ECDSA for a PQ signature



The proof system is already hash-based; only the signature check inside the circuit changes.

B. ML-DSA in the full mdoc statement

- Your ML-DSA circuit exists (858 ms on an M4, key public).
- Next: combine it with the MSO hash, the disclosed attribute and device binding, and measure end to end.
- Then the hidden-key case: ExpandA moves into the circuit.

F. A Falcon circuit for Longfellow

- Falcon verification is short: hash to a point, one polynomial product, a norm check.
- The public key is one polynomial, with nothing to expand; worth measuring when the key is hidden.

Questions for you

1. What is next for post-quantum Longfellow: issuer-side ML-DSA in mdoc, or device binding first?
2. What would make the ML-DSA circuit production-grade: speed, formal verification, or the spec?
3. Is Falcon on your radar, or is ML-DSA the only PQ target for now?
4. Where would an intern's code land: the C++ library, the Rust rewrite, or the formal side?

One more thing

I'm already in the PhD SWE intern pool.

If you think there is a fit, would you be open to requesting me from the pool for your team?

I'm happy to send a one-page project sketch this week.

Thank you!

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